"EMPIRICAL STUDY OF ORDER FLOW MICROSTRUCTURE AND SHORT-TERM PRICE IMPACT IN A CONTINUOUS DOUBLE AUCTION MARKET USING A PROPRIETARY C++ LIMIT ORDER BOOK ENGINE"

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Abstract

Purpose of the Study: *The purpose of this paper is to analyse the short-term predictive capability of the Order Imbalance (OI) metric in continuous double auction (CDA) markets. By constructing a Limit Order Book (LOB) matching engine in C++, the paper examines whether immediate volume asymmetries can reliably predict the short-term mid-price return.*

Research Methodology: *The paper adopts a system design and empirical simulation methodology. A proprietary matching engine processes simulated tick data, reconstructing the LOB state at each timestamp. The engine calculates Mid-Price and Order Imbalance at each tick, generating a proper output log for empirical analysis.*

Results of the Study:

•The C++ LOB engine successfully portrays realtime price discovery through robust order matching events.

•Order Imbalance movements are followed up by observable mid-price shifts, confirming short-term price impact.

•The simulation validates the theoretical relationship between supply-demand asymmetry and transient price movements.

•The system architecture achieves insertion and deletion complexity using nested STL(Standard Template Library) functions.

Suggestions of the Study: *It is suggested that the engine be extended to process large gigabyte real NSE tick data files. A formal Pearson Correlation Coefficient between OI_t and R_(t+10s) should be calculated. Machine learning models should be explored to build predictive alpha signals from order flow features.*

Conclusion: *The proprietary C++ LOB engine efficiently demonstrates order imbalance functions as a leading microstructural indicator of short-term price impact. The system design replicates real transactions matching logic and provides a replicable blueprint for empirical market microstructure research.*

Implication of the Study: *This paper reflects that continuous order flow analysis, grounded in robust system design, can serve as a baseline for both academic market microstructure research and practical algorithmic trading strategy developments. Implementation on real exchange data would allow robust measurement of price impact.*

Keywords: *Limit Order Book; Order Flow Imbalance; Market Microstructure; Price Impact; Continuous Double Auction; C++ Matching Engine; Algorithmic Trading.*

INTRODUCTION

In current financial markets, price formation is not a random phenomenon. Asset prices are calculated through the continuous transactions of buy and sell orders, compiled within a Limit Order Book (LOB) the core mechanism of any transaction-based Continuous Double Auction (CDA) market. Understanding how the imbalance between buying and selling pressure translates into short-term price movements is one of the foundational questions of market microstructure research.

Cont, Kukanov, and Stoikov (2014) established that the normalized order flow imbalance at the best bid and ask levels is a strong predictor of short-term mid-price changes. Their work, grounded in data from the NASDAQ exchange, demonstrated that a single imbalance variable explains a significant portion of short-term price movement. Similarly, **Gould et al. (2013)** provided a cumulative review of LOB dynamics, establishing that book state variables including order imbalance carry significant information about the movement of price movements.

The theoretical baseline of this rests in **Parlour's (1998)** model of order choice, which shows how rational agents, observing the current book state, strategically choose between market orders (robust, instant transaction) and limit orders (passive, price-improving). When buying pressure dominates, the book becomes informationally skewed: the imbalance metric approaches +1, meaning that upward price pressure will occur as assets disappear. **Foucault, Pagano, and Roell (2013)** further formalize how such order flow dynamics drive the price discovery process.

LITERATURE REVIEW

The empirical paper of order flow and its relationship to price impact has a well developed theoretical and empirical tradition.

Cont, Kukanov, and Stoikov (2014) developed one of the most cited empirical frameworks for papering order flow in LOB markets. Their core finding was that a single variable, the order flow imbalance, defined as the difference between bid-side and ask-side volumes divided by their sum is a statistically strong predictor of short-term mid-price movements. Using intraday NASDAQ data, they showed that this relationship holds across a wide range of stocks and time windows, establishing the order imbalance metric as a strong indicator of microstructural price dynamics. Their work justifies the construction of the OI metric and the short-term return prediction in the present paper.

Gould et al. (2013) provided a cumulative survey of the statistical properties and theoretical models of limit order books. They documented the empirical patterns in order insert, cancellation, and execution rates, and reviewed the a lot of structural models of LOB dynamics. They established that the state of the LOB including volume at the best bid and ask, queue lengths, and order imbalance contain forward-looking information about mid-price movements. This work justifies the approach taken in this paper: by reconstructing the full LOB state at each timestamp so that the signals can be extracted and evaluated.

Parlour (1998) developed a theoretical model of strategic order placement in limit order markets. Her model showed that order choice between limit and market orders is depends critically on the state of the existing book. When one side of the book is deep, agents on the other side face greater competition and are more likely to submit aggressive market orders, assets disappear and causing price impact. The present paper calculates this through the OI metric.

Foucault, Pagano, and Roell (2013) provided a cumulative treatment of market microstructure, mixing theoretical models with empirical evidence. Their work shows how liquidity providers and demanders interact in these markets, how unfavourable selection costs are joined in the bid-ask spread, and the degree to which incoming orders carry private information drives price impact. This frames the LOB engine made in this paper and provides the theoretical baseline for understanding OI.

METHODOLOGY

Problem Statement

While the theoretical relationship between order flow imbalance and short-term price impact is well established in the academic literature, empirical validation requires high-performance infrastructure capable of reconstructing LOB state from raw message feeds in real time. Generic high-level data analysis tools introduce latency and abstraction that obscure the microstructural dynamics. This paper addresses the question: Can a custom C++ LOB matching engine, processing simulated tick data, empirically demonstrate the relationship between Normalized Order Imbalance and 10-second forward mid-price returns?

Objectives of the Study

- To construct a high-performance C++ Limit Order Book engine capable of processing tick-by-tick order data with O(log N) complexity.
- To implement and validate core LOB operations: addOrder(), cancelOrder(), and matchOrder().
- To compute Normalized Order Imbalance (OI) and Mid-Price at each market state.
- To empirically evaluate the relationship between OI and short-term mid-price returns using simulation output.
- To establish a reproducible framework extensible to real NSE tick-by-tick data.

Data

This paper uses simulated tick data constructed to mirror the message format of a real exchange order feed. Each record contains a timestamp, order identifier, side (BUY/SELL), price, quantity, and action type (ADD/CANCEL/MATCH). The simulation covers timestamps T=1000 through T=1010, generating a sequence of order additions and aggressive crossing events designed to produce observable price impact events. The target instrument modeled is the Nifty 50 futures contract, which trades on the NSE in a continuous double auction format.

C++ SYSTEM ARCHITECTURE AND IMPLEMENTATION

Memory Architecture

The core LOB engine is built entirely from the C++ Standard Template Library (STL), deploying a nested associative container structure: std::map<double, std::map<std::string, Order>>. The outer map organizes price levels sorted in ascending order for the Ask side and in descending order (via std::greater<double>) for the Bid side, ensuring that the best bid is always accessible at begin(). This guarantees O(log N) insertions, deletions, and O(1) top-of-book lookups. The inner map indexes individual orders by their unique Order_ID string, allowing O(log M) order updates and exact execution tracking.

Core Engine Operations

The engine exposes three primary operations:

- addOrder(): Inserts a new passive limit order into the appropriate price level of the Bid or Ask book. If no price level exists, it is created dynamically. The order payload stores side, price, quantity, and a unique identifier.
- cancelOrder(): Locates the target order by iterating the price levels for the specified side and removes it from the inner map. If the price level becomes empty after removal, the level itself is erased, maintaining book cleanliness.
- matchOrder(): Simulates the exchange crossing network. Incoming aggressive orders are iteratively swept against resting limit orders at the top of the opposite book. Matching continues until the incoming quantity is exhausted or no further price cross exists. Residual unmatched volume automatically transitions into passive resting liquidity at the order's limit price.

Derived Metrics

At each timestamp, after processing the incoming message, the engine derives two key metrics:

- Mid-Price (P_mid): Calculate as the arithmetic mean of the best bid (top of Bid map) and best ask (top of Ask map), i.e., P_mid = (Best_Bid + Best_Ask) / 2. When only one side is populated, the mid-price reflects that side's best quote alone.
- Order Imbalance (OI): Calculate as OI = (V_bid - V_ask) / (V_bid + V_ask), where V_bid and V_ask are the total volumes resting at the best bid and best ask levels respectively. OI ranges in [-1, +1]: a value approaching +1 indicates demand surplus (buying pressure), while a value approaching -1 indcates supply surplus (selling pressure).

EMPIRICAL OUTPUT ANALYSIS

The engine output log for timestamps T=1000 through T=1010 provides an empirical record of microstructural dynamics. Three distinct phases are observable:

Liquidity Accumulation (T=1000 to T=1003)

At T=1000, only a BUY order exists in the book, producing an OI of +1.0, maximum demand imbalance with no supply. As sell-side orders accumulate (T=1001 to T=1003), the OI migrates toward zero and the mid-price stabilizes at 99.75. This phase confirms the engine correctly initializes the book state and calculates imbalance in the presence of one-sided liquidity.

Crossing Event (T=1003 to T=1004)

At T=1004, an incoming aggressive buy order (O5, BUY 100.5, qty 80) crosses the spread against the resting ask at 100.5, triggering: MATCH: 80 units at 100.5. The engine correctly executes the match, updates the resting ask quantities, and reflects the price discovery event in the updated book state. The OI rises from 0.0526 to 0.2245 following this execution, consistent with the depletion of ask-side supply.

Robust Buy Sweep and Price Impact (T=1008 to T=1009)

The most significant empirical event occurs at T=1009, when a large aggressive buy order (O9, BUY 101.0, qty 200) sweeps multiple layers of the ask side:

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MATCH: 10 units at 100.5
MATCH: 90 units at 100.8
MATCH: 100 units at 101.0
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This sequential depletion of supply across three consecutive price levels drives the OI from 0.0909 to 0.7143, and the mid-price immediately responds by shifting upward from 99.75 to 100.0. This event directly demonstrates the short-term price impact mechanism: aggressive order flow depletes supply, the imbalance metric spikes, and the mid-price adjusts to reflect the new informational equilibrium. This finding is consistent with the theoretical predictions of Parlour (1998) and the empirical findings of Cont, Kukanov, and Stoikov (2014).

SYSTEM OUTPUT LOG

Timestamp	Mid-Price	Order Imbalance (OI)
T=1000	0 (one-sided book)	+1.0000
T=1001	100.0	-0.2000
T=1002	100.0	+0.3333
T=1003	99.75	+0.0526
T=1004 (MATCH: 80 at 100.5)	99.75	+0.2245
T=1005	99.25	+0.0909
T=1007 (MATCH: 60 at 99.5, 30 at 100.5)	99.75	+0.3043
T=1008	99.75	+0.0909
T=1009 (MATCH: 10 at 100.5, 90 at 100.8, 100 at 101.0)	100.0	+0.7143
T=1010 (MATCH: 100 at 99.0, 50 at 98.5)	99.75	+0.5000

SIMULATED TICK DATA (CSV)

The following CSV records constitute the input feed processed by the engine:

timestamp	Order_id	side	price	quantity	type
1000	O1	BUY	99.0	100	ADD
1001	O2	SELL	101.0	150	ADD
1002	O3	BUY	98.5	200	ADD
1003	O4	SELL	100.5	120	ADD
1004	O5	BUY	100.5	80	ADD
1005	O6	SELL	99.5	60	ADD
1007	O7	BUY	100.8	90	ADD
1008	O8	SELL	100.8	90	ADD
1009	O9	BUY	101.0	200	ADD
1010	O10	SELL	98.5	150	ADD

FINDINGS OF THE STUDY

- The C++ LOB engine implements all three core operations: addOrder(), cancelOrder(), and matchOrder() with the expected algorithmic complexity, showing the viability of STL-based LOB design for high-frequency data processing.
- Order Imbalance consistently precedes and accompanies mid-price movements in the simulation output, with the most pronounced example being the OI spike from 0.0909 to 0.7143 at T=1009, coinciding with a robust sweep and an upward mid-price shift.
- The simulation confirms the theoretical prediction that aggressive order flow depletion of book liquidity is the primary mechanism of short-term price impact, consistent with the frameworks of **Cont et al. (2014)** and **Gould et al. (2013).**
- The system design is reproducible and extensible: the engine architecture can be adapted to process real NSE tick data with minimal modification.
- A formal Pearson Correlation Coefficient between OI_t and R_(t+10s) has not yet been computed, representing the primary empirical limitation of the current paper stage.

LIMITATIONS OF THE STUDY

- The paper uses simulated tick data rather than real NSE exchange data. While the simulation is designed to replicate realistic order arrival patterns, it does not show the full statistical distribution of order sizes, arrival rates, and cancellation patterns observed in live markets.
- The Pearson Correlation Coefficient between OI_t and R_(t+10s) has not been formally computed in this paper. Without this statistical measure, the quantitative strength of the OI-to-price-impact relationship cannot be precisely evaluated.
- This engine implements a imbalance metric rather than a pure Level-1 imbalance.

- The paper does not account for adverse selection costs, market impact decay, or the distinction between informed and uninformed order flow dimensions that are central to a full empirical treatment of market microstructure.

CONCLUSION

This paper demonstrates that a custom C++ Limit Order Book engine, built from STL, can successfully replicate the core mechanics of a continuous double auction exchange and generate empirically meaningful microstructural output. The engine's real-time computation of Order Imbalance and Mid-Price, applied to simulated tick data, reveals a clear and consistent pattern: aggressive order flow that depletes one side of the book produces immediate, observable price impact a finding that aligns with the theoretical frameworks of **Parlour (1998)** and the empirical evidence of **Cont, Kukanov, and Stoikov (2014)**.

The next phase of this research involves deploying the engine against multi-gigabyte real NSE tick data to compute the formal Pearson Correlation Coefficient between OI_t and R_(t+10s), providing a statistically rigorous quantification of the signal's predictive power. If confirmed, this would establish order flow imbalance as a viable high-frequency alpha signal in the Indian equity derivatives market.

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